

API Documentation

Overview: The AgroChill API provides real-time ingestion, forecasting, and evaluation services for crop price prediction using XGBoost models. It supports weather and price data ingestion, incremental training, model evaluation, and future price forecasting.

Endpoints:

1. POST /api/data/weather

Ingest new weather data and trigger incremental training.

Request Body: List[WeatherRecord]

Response:

```
{ "status": "success", "ingested": 1, "rmse": 2.57, "nrmse": 0.195 }
```

2. GET /api/data/weather?region=Arcadia

Retrieve the latest weather data for a given region.

Query Parameter:

region: str

Response:

```
{ "status": "success", "weather data": [ { "date": "2044-01-07", "region": "Arcadia",  
"temperature": 298.15, "rainfall": 10.2, "humidity": 68.0, "yield_impact": 1.1 } ] }
```

3. POST /api/data/prices

Ingest new price data and trigger incremental training for relevant commodities.

Request Body: List[PriceRecord]

Example:

```
[ { "date": "2044-01-07", "region": "Arcadia", "commodity": "Plantain", "price": 40.0,  
"type": "Fruit" } ]
```

Response:

```
{ "status": "success", "ingested": 1, "evaluations": [ { "commodity": "Plantain", "rmse":  
3.45, "nrmse": 0.234 } ] }
```

4. GET /api/data/price?commodity=Plantain®ion=Arcadia

Retrieve the latest price records for a commodity-region combination.

Query Parameters:

commodity: str

region: str

Response:

```
{ "status": "success", "price data": [ { "date": "2044-01-07", "region": "Arcadia",  
"commodity": "Plantain", "price": 40.0, "type": "Fruit" } ] }
```

5. POST /api/predict

Return a rolling 4-week price forecast for a given commodity and region.

Query Parameters (or JSON body if preferred):

commodity: str

region: str

Response:

```
{ "commodity": "Plantain", "region": "Arcadia", "forecast": [ { "date": "2044-01-14",  
"predicted_price": 41.5}, { "date": "2044-01-21", "predicted_price": 42.1}, { "date":  
"2044-01-28", "predicted_price": 43.0}, { "date": "2044-02-04", "predicted_price": 42.7}  
], "rmse": 3.1, "nrmse": 0.182 }
```

6. POST /api/evaluate

Evaluate all models and return overall RMSE and NRMSE.

Response:

```
{ "status": "success", "rmse": 3.28, "nrmse": 0.201, "evaluations": [ { "commodity":  
"Plantain", "RMSE": 3.1, "NRMSE": 0.182 }, { "commodity": "Cabbage", "RMSE": 2.7,  
"NRMSE": 0.145 } ] }
```

7. POST /api/evaluate_model

Evaluate a specific commodity model.

Query Parameter:

commodity: str

Response:

```
{ "status": "success", "commodity": "Plantain", "rmse": 3.1, "nrmse": 0.182 }
```

8. POST /api/train_scratch

Retrain all models from scratch using the full training dataset.

Response:

```
{ "status": "success" }
```